

Day 3 — Detailed Study & Lab Guide

Hybrid Identity: On-Prem Active Directory to Microsoft Entra ID — plain-language concepts, real architecture, exact navigation, hands-on labs, troubleshooting, and interview prep.

LAB VALUES USED THROUGHOUT THIS GUIDE

- Public domain (GoDaddy): **minjtech.xyz**
- AD DNS domain (recommended): **ad.minjtech.xyz** — UPN suffix added: **minjtech.xyz**
- Resource Group: **rg-minjtech-hybrid-id** | VNet: **vnet-minjtech-hybrid (10.10.0.0/16)**
- DC01 = Domain Controller (10.10.1.10) | AADCONNECT01 = Entra Connect server (10.10.1.20) | CLIENT01 = test client (10.10.2.10)
- All IPs/names are LAB VALUES — change them to match your own environment.

Section 1: What Is Hybrid Identity? (In Plain Words)

CONCEPT — In Simple Words

Imagine your company already has an **on-prem Active Directory** — a local employee register running on a Windows Server, used for logging into office PCs for the last 15 years. Now the company also wants to use Microsoft 365, Teams, and Azure — all of which live in the **cloud** and use **Microsoft Entra ID** (formerly Azure AD) as their identity system.

The question: should employees have TWO separate accounts — one for the office PC, one for the cloud apps — with two different passwords? That's confusing and insecure (people write passwords on sticky notes!). **Hybrid Identity** solves this: it keeps on-prem AD as the "source of truth," and automatically copies (synchronizes) the same user accounts into Entra ID — so ONE identity works everywhere.

- **On-prem Active Directory** — the local, company-owned directory (runs on your own Windows Server).
- **Microsoft Entra ID** — Microsoft's cloud-based identity directory (multi-tenant, internet-facing).
- **Microsoft Entra Connect** — the bridge/sync tool that copies users, groups, and password hashes from on-prem AD into Entra ID, continuously, in the background.
- **Identity Synchronization vs Authentication** — sync is COPYING the account data one-way (AD → Entra ID); authentication is the act of PROVING who you are at sign-in time. Sync happens every few minutes in the background; authentication happens every time someone logs in.
- We use **PHS (Password Hash Sync)** in this lab because it's the simplest, most resilient method — Entra ID can authenticate users even if the on-prem network or DC01 is temporarily down.

Hybrid Identity — The Big Picture

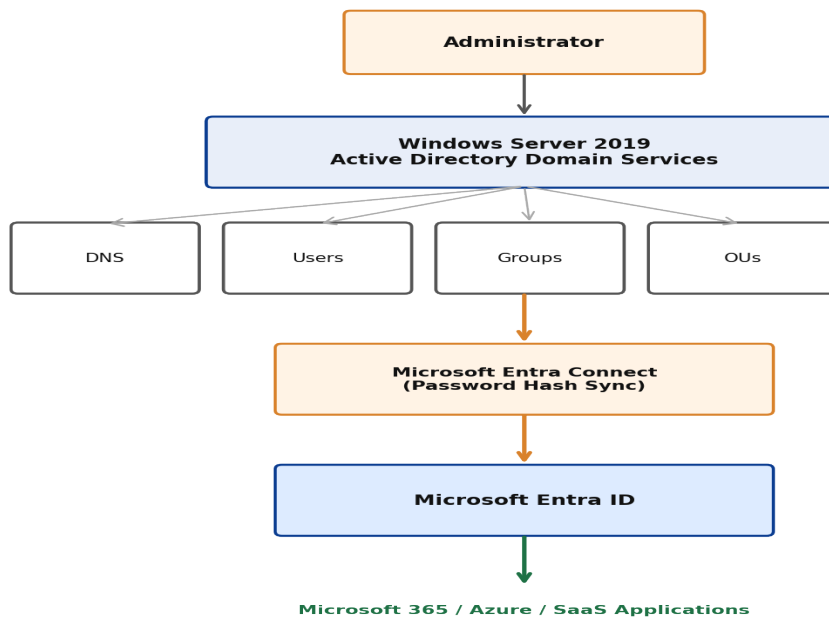


Diagram 1 — the complete hybrid identity chain: AD → Entra Connect → Entra ID → cloud apps.

Section 2: Lab Architecture

CONCEPT — In Simple Words

We are going to build a mini version of a real company network, entirely inside Azure — using Azure VMs to simulate "on-prem" servers. This is a common and legitimate way to LAB and LEARN hybrid identity without needing physical hardware.

- **DC01** — Windows Server 2019 VM that becomes our Domain Controller, running AD DS + DNS.
- **AADCONNECT01** — a SEPARATE Windows Server 2019 VM that runs Microsoft Entra Connect (kept separate from DC01 deliberately — see security section for why).
- **CLIENT01** (optional) — a Windows 10/11 VM we domain-join, to test real user sign-in and SSO.
- All three sit inside one VNet so they can talk to each other privately, with DNS pointed at DC01.

Lab Network Architecture (example private IPs — change as needed)

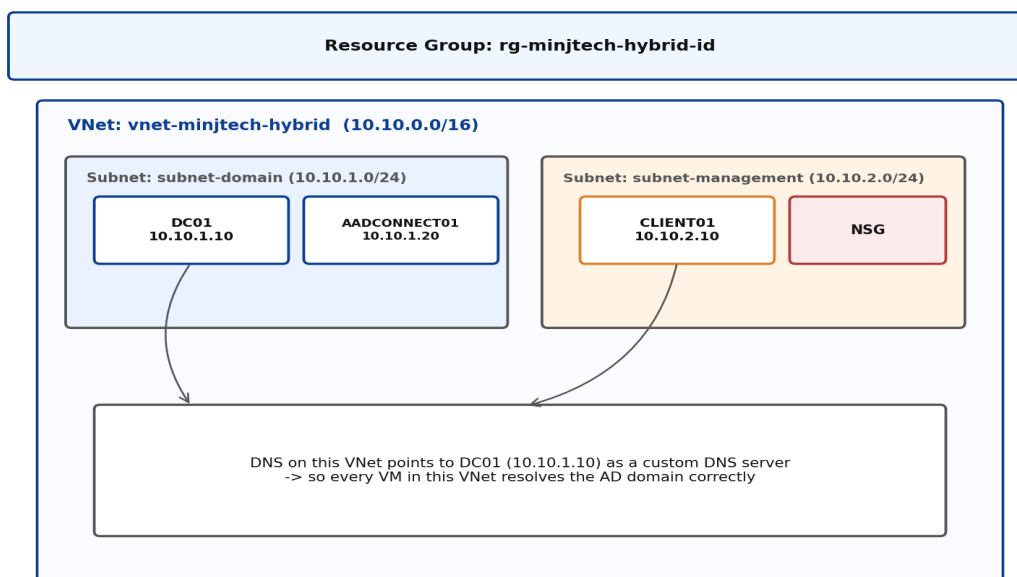


Diagram 2 — Resource Group → VNet → two subnets → three VMs, all example lab values.

PORTAL NAVIGATION — Create the Network Foundation

- 1 portal.azure.com → search **Resource groups** → + Create → name **rg-minjtech-hybrid-id** → pick a region → Review + create
- 2 Search **Virtual networks** → + Create → Resource group = rg-minjtech-hybrid-id → Name = **vnet-minjtech-hybrid** → IPv4 address space = 10.10.0.0/16
- 3 On the IP Addresses tab → + Add subnet → name **subnet-domain**, range 10.10.1.0/24 → Add another subnet **subnet-management**, range 10.10.2.0/24 → Review + create
- 4 Search **Network security groups** → + Create → name **nsg-minjtech-hybrid** → associate it to subnet-management (client subnet) after creation

MINI-LAB: Stand Up the Network

- 1 Create the Resource Group, VNet, and both subnets exactly as above

- 2 Create the NSG and confirm it shows under the VNet's subnet-management → Network security group blade
- 3 Note down the VNet's address space and both subnet ranges in your lab notebook — you'll assign static IPs from these ranges next

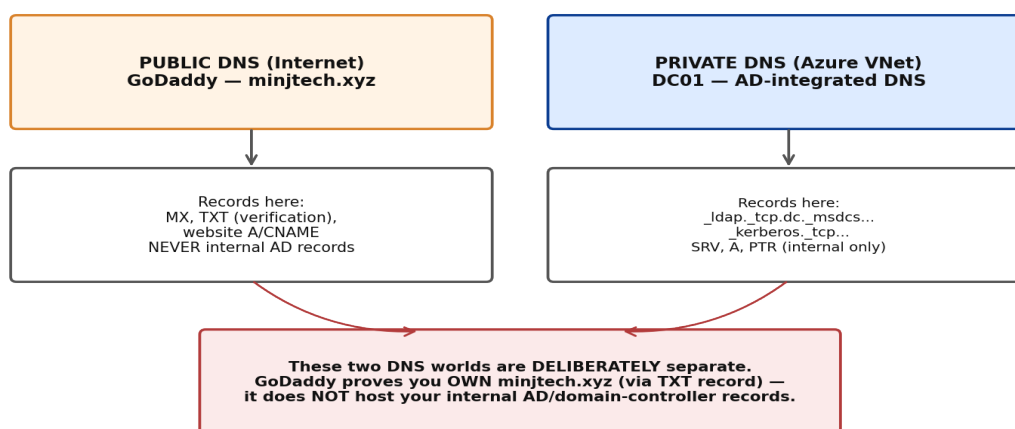
Section 3: Public DNS (GoDaddy) vs Private AD DNS

CONCEPT — In Simple Words

You already own **minjtech.xyz** through GoDaddy. It's tempting to think "I'll just put all my AD records there too" — but that would be a serious mistake. Public DNS (GoDaddy) is for things the **WHOLE INTERNET** needs to find (your website, your email server). Private AD DNS (hosted on DC01) is for things **ONLY** your internal network needs to find (domain controllers, LDAP, Kerberos). Mixing them exposes your internal infrastructure to the internet — a security risk.

- **A record** — maps a name to an IPv4 address (e.g. `www.minjtech.xyz` → `20.1.2.3`).
- **CNAME** — an alias, one name pointing to another name.
- **NS record** — tells the internet which DNS server is authoritative for a domain.
- **SRV record** — tells clients **WHERE** a specific service lives, with priority/weight/port (this is how AD is found — e.g. "where's the nearest domain controller for LDAP?").
- **TXT record** — free-text record, commonly used to **PROVE** domain ownership (this is how Entra ID verifies `minjtech.xyz` belongs to you).
- AD automatically creates records like `_ldap._tcp.dc._msdcs.ad.minjtech.xyz` and `_kerberos._tcp.ad.minjtech.xyz` — these SRV records let domain-joined machines locate a domain controller automatically. They live **ONLY** in your private AD DNS zone, never in GoDaddy.

Public DNS vs Private AD DNS — Keep Them Separate



Never expose Domain Controller ports (LDAP 389, Kerberos 88, DNS 53) to the public Internet

Diagram 3 — public GoDaddy DNS and private AD DNS solve two completely different problems.

GODADDY NAVIGATION (for domain verification later)

- 1 GoDaddy → My Products → Domains → minjtech.xyz → DNS → Manage DNS
- 2 This is where you will **ONLY** add: the TXT record Entra ID gives you for domain verification (Section 8), and optionally MX records if you route mail through Microsoft 365 later
- 3 Do NOT add A records for DC01, AADCONNECT01, or any internal server here

Section 4: Domain Naming Strategy

CONCEPT — In Simple Words

What should we actually NAME our Active Directory domain? This decision is hard to undo later, so let's compare the options plainly:

- **minjtech.xyz** (same as public domain) — simplest to look at, but risky: your internal AD domain name becomes identical to your internet-facing name, which can cause DNS resolution conflicts and makes it harder to keep public/private DNS cleanly separate.
- **ad.minjtech.xyz** (a subdomain) — RECOMMENDED. Internal AD lives at ad.minjtech.xyz, clearly separated from the public zone, while users still log in with clean emails like nishant@minjtech.xyz using a UPN suffix (explained below).
- **corp.minjtech.xyz** — same idea as ad.minjtech.xyz, just a different naming preference. Equally valid.
- **minjtech.local** — the OLD-STYLE approach (non-routable .local domain). AVOID for new builds: .local isn't a real internet domain, causes certificate and Entra Connect complications, and Microsoft no longer recommends it.

Decision for this lab: AD domain = **ad.minjtech.xyz**. We then add **minjtech.xyz** as an additional UPN suffix so a user's actual AD domain (ad.minjtech.xyz) can be different from their login email-style username (nishant@minjtech.xyz) — best of both worlds.

- **AD Forest** — the outermost security/replication boundary; one forest can have multiple domains.
- **AD Domain** — one namespace within the forest (ad.minjtech.xyz in our lab).
- **UPN suffix** — the part after @ in a login name; can be different from the actual AD domain name.
- **Public domain** — minjtech.xyz, owned via GoDaddy, used for the internet-facing identity (email-style logins, Entra ID custom domain).
- **Entra ID domain** — once minjtech.xyz is verified in Entra ID, cloud accounts can also use it as their UPN, matching on-prem exactly.

Section 5: Build DC01 & Install Active Directory Domain Services

CONCEPT — In Simple Words

DC01 is going to be the "brain" of our identity system — the server that stores every user, password hash, and group. We give it a fixed (static) private IP so it never moves, since every other machine will be told "go ask THIS IP for directory/DNS answers." Installing **AD DS** (Active Directory Domain Services) turns a plain Windows Server into a Domain Controller.

PORTAL NAVIGATION — Create the VM

- 1 Search **Virtual machines** → + Create → Azure virtual machine
- 2 Resource group = rg-minjtech-hybrid-id, VM name = **DC01**, Region = same as VNet, Image = **Windows Server 2019 Datacenter**
- 3 Size: a small/medium size is fine for a lab (e.g. Standard_B2s or D2s_v3)
- 4 Set Administrator username/password (save it securely — this becomes your first Domain Admin login)
- 5 Networking tab → Virtual network = vnet-minjtech-hybrid, Subnet = subnet-domain, Public IP = None (or Basic only if you need temporary RDP — remove after setup, see Security section)
- 6 NIC → after VM creation, go to DC01 → Networking → click the network interface → IP configurations → change Private IP allocation to **Static**, set it to **10.10.1.10**
- 7 Review + create → Create → wait for deployment

CONNECT & CONFIGURE WINDOWS SERVER

- 1 DC01 Overview → Connect → RDP → download RDP file → sign in with the admin credentials you set
- 2 Server Manager opens automatically → Local Server → click the computer name → rename to **DC01** if not already, restart if prompted
- 3 Local Server → set the correct Time zone (important for Kerberos, which is time-sensitive)
- 4 Confirm Windows Update is current (Settings → Windows Update)

```
# PowerShell verification commands
hostname
ipconfig /all
Get-NetIPConfiguration
Get-NetIPAddress
```

hostname confirms the computer name. **ipconfig /all** and the **Get-NetIP*** cmdlets confirm the static private IP (10.10.1.10) is actually applied inside Windows, not just in the Azure portal.

INSTALL ACTIVE DIRECTORY DOMAIN SERVICES

- 1 Server Manager → Manage → Add Roles and Features → Next past the intro
- 2 Installation type = Role-based or feature-based installation → Next
- 3 Select DC01 from the server pool → Next
- 4 Server Roles → check **Active Directory Domain Services** → a popup asks to add required features → click **Add Features** → Next through DNS Server role (it gets added automatically too) → Next → Install
- 5 After install completes, click the yellow flag notification → **Promote this server to a domain controller**

PROMOTE TO DOMAIN CONTROLLER — WIZARD SCREENS

- 1 Deployment Configuration → select **Add a new forest** → Root domain name = **ad.minjtech.xyz**
- 2 Domain Controller Options → Forest functional level and Domain functional level = Windows Server 2016 (or highest available) → confirm **DNS server** is checked (installs DNS role alongside AD) → set a **Directory Services Restore Mode (DSRM) password** — save this separately, it's used for disaster recovery boot mode
- 3 DNS Options → you'll see a warning about not being able to create a delegation — this is NORMAL and expected in a lab, click Next
- 4 Additional Options → confirm the NetBIOS name (auto-generated, e.g. AD) looks right → Next
- 5 Paths → leave default paths for the **Database folder**, **Log files folder**, and **SYSVOL folder** unless you have a specific reason to change them → Next
- 6 Review Options → Next → Prerequisites Check → if all checks pass (warnings are usually OK, errors are not) → Install
- 7 The server automatically reboots when finished — you're now logging into a real Domain Controller

Lab Network Architecture (example private IPs — change as needed)

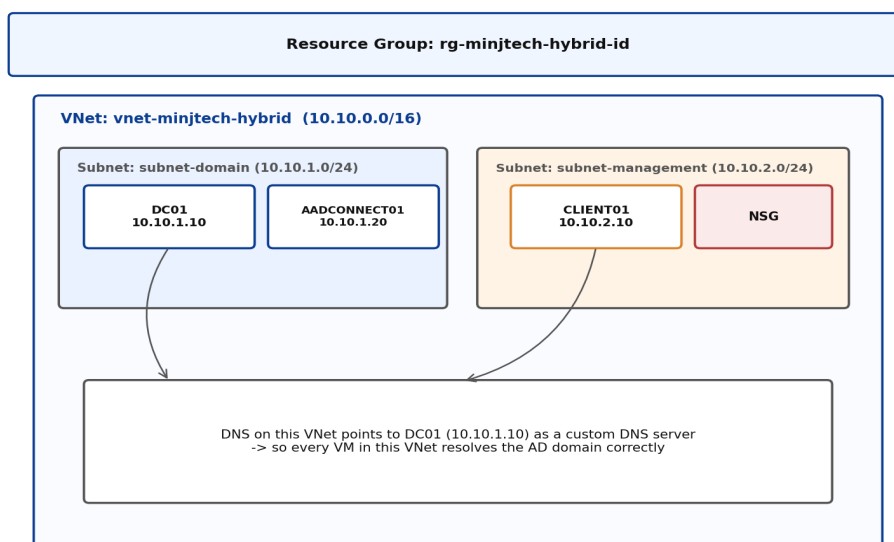


Diagram 4 (reuse) — DC01 now runs both AD DS and DNS at 10.10.1.10 inside subnet-domain.

MINI-LAB: Build DC01 End-to-End

- 1 Create DC01 with a static private IP 10.10.1.10
- 2 Install the AD DS + DNS roles
- 3 Promote it to create a NEW FOREST: ad.minjtech.xyz
- 4 After reboot, log back in and confirm the login screen now shows the domain name
- 5 Run **Get-ADDomain** and **Get-ADForest** in PowerShell and confirm the output matches ad.minjtech.xyz

Section 6: Verify AD, Configure DNS & UPN Suffix

CONCEPT — In Simple Words

Right after promotion, AD DS silently created a lot of internal structure: **NTDS.dit** (the actual database file holding every object), **SYSVOL** (a shared folder replicating policies/scripts between domain controllers), and DNS zones with special **SRV records** so any computer can automatically "discover" a domain controller. We now confirm all of this is healthy before building anything on top of it.

- **NTDS.dit** — the database file that stores every user, group, and computer object.
- **SYSVOL** — a special replicated folder holding Group Policy and login scripts.
- **Kerberos** — the protocol used for domain authentication tickets (time-sensitive, hence the time zone setting earlier).
- **LDAP** — the protocol used to read/write directory objects (users, groups).
- **Global Catalog** — a searchable index of objects across the whole forest.
- **FSMO roles** — five special jobs (like "who assigns the next available ID") held by specific domain controllers — with only one DC, DC01 holds all five automatically.

VERIFY AD — COMMANDS & GUI

```
dcdiag
dcdiag /test:dns
Get-ADDomain
Get-ADForest
Get-ADUser -Filter *
Get-ADComputer -Filter *
Get-ADGroup -Filter *
```

dcdiag runs a full built-in health check of the domain controller — always run this after any AD change. **dcdiag /test:dns** isolates DNS-specific problems, which cause the majority of real-world AD issues.

- 1 GUI verification: Server Manager → Tools → **Active Directory Users and Computers** — you should see the ad.minjtech.xyz domain with default containers (Users, Computers, Domain Controllers)
- 2 Server Manager → Tools → **DNS Manager** → expand DC01 → Forward Lookup Zones → ad.minjtech.xyz → confirm folders like `_msdcs`, `_sites`, `_tcp`, `_udp` exist (these hold the SRV records)
- 3 Inside `_tcp`, confirm you see entries like `_ldap` and `_kerberos`

```
_ldap._tcp.dc._msdcs.ad.minjtech.xyz
_kerberos._tcp.ad.minjtech.xyz
```

DNS COMMANDS

```
nslookup ad.minjtech.xyz
Resolve-DnsName ad.minjtech.xyz
ipconfig /flushdns
ipconfig /registerdns
```

nslookup/Resolve-DnsName prove the domain resolves to DC01's IP. **ipconfig /flushdns** clears a machine's local DNS cache (useful after making DNS changes). **ipconfig /registerdns** forces a machine to re-register its own DNS record — handy if a machine seems "missing" from DNS.

CONFIGURE UPN SUFFIX

- 1 Server Manager → Tools → Active Directory Domains and Trusts → right-click the root → **Properties**
- 2 UPN Suffixes tab → type **minjtech.xyz** → click Add → OK
- 3 Now when creating a user, the UPN dropdown will offer BOTH ad.minjtech.xyz and minjtech.xyz — always choose **minjtech.xyz** for real users, so their login looks like a normal email address

Example: username **nishant**, AD login becomes **nishant@minjtech.xyz** (clean, email-style) even though the underlying AD domain is technically ad.minjtech.xyz. This UPN match is exactly what lets Entra Connect line up the on-prem account with the Entra ID account seamlessly later.

Section 7: Create OUs, Users & Groups

CONCEPT — In Simple Words

An **Organizational Unit (OU)** is just a folder inside AD used to organize objects and apply Group Policy or (later) sync filtering. Important distinction: **OUs are NOT a security boundary** — putting a user in an OU does not by itself restrict who can log in as them; that's what Security Groups and permissions are for. OUs are about ORGANIZATION, Groups are about ACCESS.

We create a dedicated **SyncUsers** OU because later, Entra Connect will be configured with **OU filtering** — only synchronizing accounts sitting inside specific OUs, keeping ServiceAccounts and Computers out of the cloud directory entirely.

NAVIGATION — Create the OU Structure

- 1 Active Directory Users and Computers → right-click ad.minjtech.xyz → New → Organizational Unit → name = **MinjTech**
- 2 Right-click MinjTech → New → Organizational Unit → repeat for: **Users, Groups, Computers, ServiceAccounts, Admins, SyncUsers** (each nested inside MinjTech)

NAVIGATION — Create Test Users

- 1 Right-click OU=SyncUsers → New → User
- 2 Fill: First name, Last name, **User logon name** (e.g. nishant) — pick **@minjtech.xyz** from the UPN dropdown
- 3 Set a password, UNCHECK "User must change password at next logon" for lab simplicity (in production, leave it checked), CHECK "Password never expires" for lab test accounts only
- 4 Repeat for: **testuser01, testuser02, cloudadmin**
 - **User logon name** — the sAMAccountName / pre-Windows 2000 login name (short form).
 - **UPN** — user@domain, used for modern logins and Entra ID matching.
 - **Password expiration** — a real security control; disable ONLY for lab/demo accounts.

NAVIGATION — Create Groups

- 1 Right-click OU=Groups → New → Group → Name = **GG-Hybrid-Users**, Group scope = Global, Group type = Security
- 2 Repeat for **GG-DevOps** and **GG-CloudUsers**
- 3 Double-click a group → Members tab → Add → type user names → Check Names → OK to add nishant/testuser01/etc.

MINI-LAB: Populate the Directory

- 1 Build the full OU tree under MinjTech exactly as listed above
- 2 Create all 4 test users inside OU=SyncUsers with the minjtech.xyz UPN
- 3 Create all 3 groups inside OU=Groups and add at least 2 users to each
- 4 Run **Get-ADUser -Filter * -SearchBase "OU=SyncUsers,OU=MinjTech,DC=ad,DC=minjtech,DC=xyz"** and confirm all 4 users appear

Section 8: Prepare Microsoft Entra ID & Verify minjtech.xyz

CONCEPT — In Simple Words

Every Azure subscription is linked to a **Tenant** — your organization's dedicated instance of Entra ID (e.g. yourcompany.onmicrosoft.com by default). Before Entra Connect can make your users look like nishant@minjtech.xyz in the cloud, Entra ID must first be told "minjtech.xyz really does belong to this tenant" — and the only way to prove that is by adding a special DNS record that only the domain owner could add.

- **Tenant / Tenant ID** — your org's unique identity container in Entra ID.
- **Custom domain** — a domain you own, added and verified so accounts can use it instead of the default onmicrosoft.com name.
- **Global Administrator** — full control over the tenant (needed to add/verify domains).
- **Hybrid Identity Administrator** — a scoped role specifically for managing Entra Connect and hybrid identity configuration (least-privilege alternative to Global Admin for day-2 operations).

NAVIGATION — Add & Verify the Custom Domain

- 1 portal.azure.com → search **Microsoft Entra ID** → left menu → **Custom domain names** (previously called "Domain names") → **+ Add custom domain**
- 2 Type **minjtech.xyz** → Add — Entra ID now generates a unique TXT verification value for YOUR tenant (do not use any example value from elsewhere — always copy the real one shown to you)
- 3 Copy that TXT record's Name/Host and Value/Points to exactly as shown

GODADDY: Add the TXT Verification Record

- 1 GoDaddy → My Products → Domains → minjtech.xyz → DNS → Manage DNS
- 2 Add Record → Type = TXT → Host = @ (root) → Value = **[LAB VALUE — paste the exact TXT value Entra ID generated for your tenant]** → TTL = default → Save
- 3 Wait for DNS propagation (can take a few minutes up to a few hours)
- 4 Back in Entra ID → Custom domain names → click minjtech.xyz → **Verify**

LAB VALUE vs PRODUCTION: the TXT value is unique per tenant and generated dynamically by Microsoft — never reuse a value from a tutorial or another tenant.

Section 9: Build AADCONNECT01 & Join It to the Domain

CONCEPT — In Simple Words

Why a SEPARATE server from DC01? Two reasons: (1) Best practice — never install extra software on a Domain Controller if you can avoid it, to keep its attack surface minimal; (2) If Entra Connect crashes or needs a reboot for updates, your domain controller (and everyone's ability to log into office PCs) isn't affected at all.

PORTAL NAVIGATION — Create the VM

- 1 Virtual machines → + Create → Resource group = rg-minjtech-hybrid-id, Name = **AADCONNECT01**, Image = Windows Server 2019 Datacenter
- 2 Networking → VNet = vnet-minjtech-hybrid, Subnet = subnet-domain, Public IP = None
- 3 After creation → Networking → NIC → set static private IP **10.10.1.20**
- 4 Networking → confirm the VNet's DNS settings point to DC01 (10.10.1.10) — check VNet → DNS servers → Custom → 10.10.1.10. Without this, AADCONNECT01 can never find the AD domain.

JOIN TO DOMAIN

- 1 RDP into AADCONNECT01 → Server Manager → Local Server → click Workgroup
- 2 System Properties → Change... → Member of: Domain → type **ad.minjtech.xyz** → OK
- 3 Enter Domain Admin credentials when prompted → OK → restart when prompted
- 4 Log back in using **ad.minjtech.xyz\yourdomainadmin** (or nishant@minjtech.xyz if given rights) to confirm domain join worked

```
whoami
systeminfo
Get-CimInstance Win32_ComputerSystem
```

whoami should now show DOMAIN\username instead of a local account. **Get-CimInstance Win32_ComputerSystem** confirms the Domain property equals ad.minjtech.xyz.

VERIFY CONNECTIVITY TO DC01 (before installing Entra Connect)

```
nslookup minjtech.xyz
nslookup ad.minjtech.xyz
Test-NetConnection DC01 -Port 389
Test-NetConnection DC01 -Port 636
Test-NetConnection DC01 -Port 88
Test-NetConnection DC01 -Port 53
```

- **Port 389** — LDAP (reads/writes AD objects, unencrypted).
- **Port 636** — LDAPS (LDAP over SSL/TLS, encrypted).
- **Port 88** — Kerberos authentication.
- **Port 53** — DNS lookups.

If any Test-NetConnection returns False, STOP — fix DNS/NSG/firewall before continuing to install Entra Connect, or the install will fail or silently misbehave later.

Section 10: Install Microsoft Entra Connect & Configure PHS

CONCEPT — In Simple Words

"Azure AD Connect" was RENAMED to **Microsoft Entra Connect** — same tool, new name, following Microsoft's Azure AD → Entra ID rebrand. During install, we must choose HOW passwords will be validated in the cloud. This lab uses **Password Hash Synchronization (PHS)** — explained fully below.

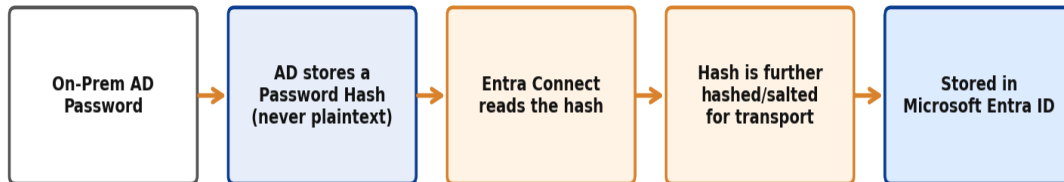
PREREQUISITES

- AADCONNECT01 is domain-joined (done in Section 9)
- An Entra ID **Global Administrator** (or Hybrid Identity Administrator) account and password ready
- An on-prem **Enterprise Admin** (or equivalent delegated rights) account ready
- minjtech.xyz already verified in Entra ID (Section 8)
- Time is in sync between AADCONNECT01 and DC01 (Kerberos is time-sensitive)

INSTALLATION STEPS

- 1 On AADCONNECT01, download **Microsoft Entra Connect** from Microsoft's official site → run the installer
- 2 Accept license terms → Continue
- 3 Choose **Customize** (NOT Express Settings) — Express Settings assumes PHS + syncs ALL of AD, but we specifically want OU filtering, so Customize gives us control
- 4 Install required components → Next (default paths are fine for a lab)
- 5 User sign-in page → select **Password Hash Synchronization** → Next
- 6 Connect to Microsoft Entra ID → sign in with your Global Administrator cloud credentials
- 7 Connect your directories → sign in with your on-prem Enterprise Admin credentials → Add Directory
- 8 Domain/OU filtering screen → choose **Sync selected domains and OUs** → expand ad.minjtech.xyz → check ONLY: MinjTech > Users, MinjTech > SyncUsers, MinjTech > Groups (leave Computers and ServiceAccounts UNCHECKED)
- 9 Uniquely identifying users → leave default (uses AD's ObjectGUID as the Source Anchor — recommended)
- 10 Filter users and devices / Optional features → for this lab, leave optional features default (Password writeback and others can be explored later)
- 11 Ready to configure → confirm **Start the synchronization process when configuration completes** is checked → Install

Password Hash Synchronization (PHS) — What Actually Travels



The PLAINTEXT password is NEVER sent to or stored in Microsoft Entra ID — only a hash of a hash.

Diagram 5 — PHS travels a hash of a hash; plaintext passwords never leave the on-prem network.

PHS vs PTA vs ADFS — COMPARISON

Feature	PHS	PTA	ADFS
Password validation	In the cloud (hash compare)	On-prem, in real time via lightweight agent	On-prem, via federation server
On-prem dependency	None after sync (cloud can auth even if on-prem is down)	High — on-prem agent must be online for every login	Very high — ADFS farm must be online for every login
Infrastructure	Just Entra Connect	Entra Connect + PTA agent(s)	ADFS servers + WAP proxy servers
Complexity	Low	Medium	High
Availability	High (cloud-native)	Depends on agent uptime/redundancy	Depends on ADFS farm uptime
Cloud authentication	Yes, fully cloud-based	Cloud-brokered, on-prem validated	Fully on-prem validated, cloud redirects
Typical use case	Most orgs — simplest, resilient, recommended default	Orgs with strict "password never leaves on-prem" policy	Orgs needing complex custom auth (smart cards, legacy federation, strict compliance)

Why this lab uses PHS: it's the simplest to deploy, requires the least ongoing infrastructure, and keeps working even if AADCONNECT01 or DC01 goes down temporarily — ideal for learning and for most real-world small/medium deployments.

Section 11: OU Filtering, the Sync Engine & First Synchronization

CONCEPT — In Simple Words

We already told the installer which OUs to sync (Section 10). Let's understand WHY and HOW that works under the hood. Entra Connect doesn't just "copy" data blindly — it runs a proper pipeline: it **imports** raw data from AD into a local staging area (Connector Space), merges everything into a central working copy (the **Metaverse**), applies rules, then **exports** the result out to Entra ID.

- **Connector** — a plug-in that talks to one specific system (one connector for on-prem AD, one for Entra ID).
- **Connector Space** — a local cache of what that connector last saw.
- **Metaverse** — the central, de-duplicated view combining data from all connectors — this is where join/merge logic decides "this AD user and this cloud user are the same person."
- **Import** → **Synchronize** → **Export** — the three-stage pipeline that runs every sync cycle.

Entra Connect Sync Engine — Import -> Metaverse -> Export

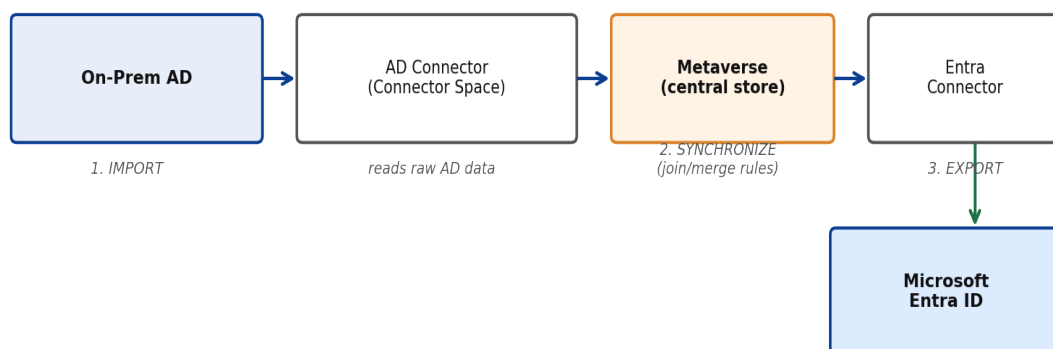
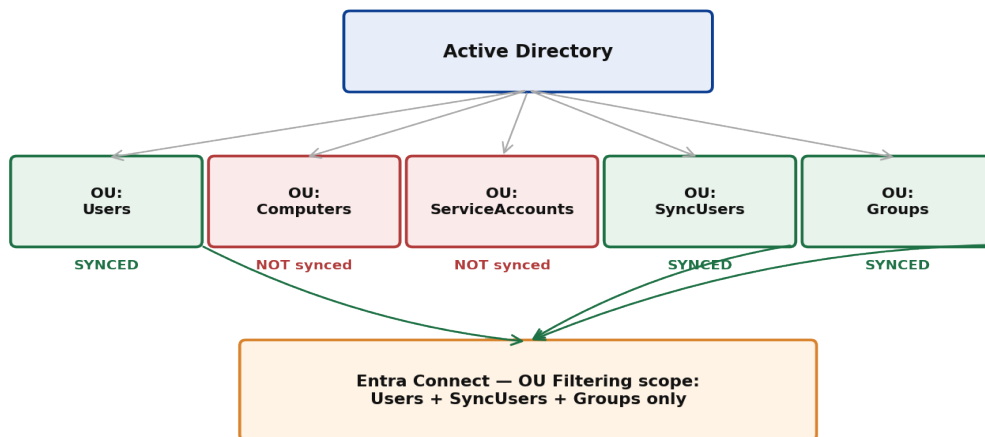


Diagram 6 — the 3-stage sync pipeline every Entra Connect cycle runs through.

OU Filtering — Sync Only What You Need



Computers and ServiceAccounts OUs are excluded on purpose — smaller, cleaner cloud directory

Diagram 7 — only Users, SyncUsers, and Groups OUs are in scope; Computers and ServiceAccounts are excluded.

RUN THE FIRST SYNC

```
Start-ADSyncSyncCycle -PolicyType Initial
Get-ADSyncScheduler
```

-PolicyType Initial forces a FULL sync (reads everything again), as opposed to **Delta** (only changes since last run) which is what runs automatically every 30 minutes by default. **Get-ADSyncScheduler** shows you the current schedule and confirms sync is enabled.

Expected timing: a small lab directory (a handful of users) typically completes an initial sync within 1–2 minutes. Larger directories take longer proportionally.

Section 12: Verify the User in Entra ID & Attribute Flow

PORTAL NAVIGATION

- 1 portal.azure.com → Microsoft Entra ID → left menu → Users → **All users**
- 2 Search for **nishant@minjtech.xyz** → click to open the user profile
 - **User principal name** — should read nishant@minjtech.xyz, matching the on-prem UPN exactly.
 - **Source** — should show **Windows Server AD** (proves this account is synced, not cloud-native).
 - **Object ID** — Entra ID's own internal unique identifier for this object.
 - **On-premises sync enabled** — should show Yes.
 - **On-premises domain name** — should show ad.minjtech.xyz.
 - **Last sync** — timestamp confirming when this object last synced successfully.

ATTRIBUTE FLOW — WHAT COPIES WHERE

```
AD (on-prem)  Entra ID (cloud)
-----
sAMAccountName = nishant --> (used internally, not shown as UPN)
userPrincipalName = nishant@minjtech.xyz --> UserPrincipalName = nishant@minjtech.xyz
displayName = Nishant Minj --> Display Name = Nishant Minj
mail = nishant@minjtech.xyz --> Mail = nishant@minjtech.xyz
objectGUID = {xxxxxxxx-...} --> Used as Source Anchor / immutableId
```

The **objectGUID** becomes the permanent link between the on-prem and cloud copies of this identity — this is why deleting and recreating an AD user (instead of just renaming it) breaks the link and creates a brand NEW Entra ID object rather than updating the existing one.

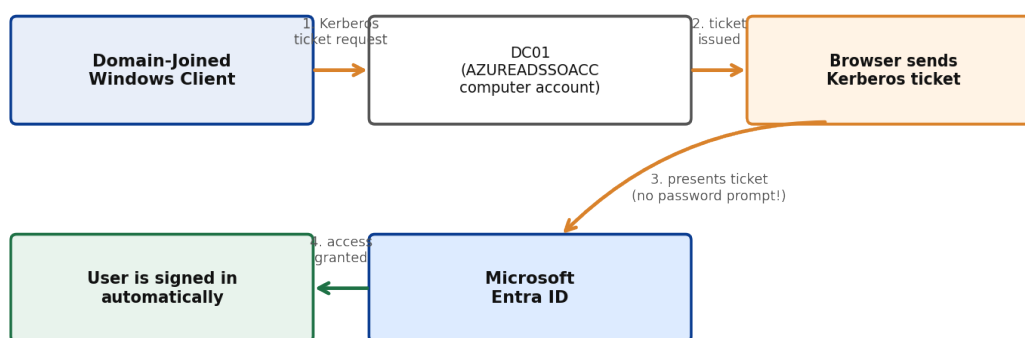
Section 13: Seamless SSO & Windows Client Test

CONCEPT — In Simple Words

PHS alone means a user still has to TYPE their password when opening a cloud app — Entra ID just checks it against the synced hash. **Seamless SSO** goes one step further: if the user is already logged into a DOMAIN-JOINED Windows PC on the corporate network, Entra ID can sign them in AUTOMATICALLY using a Kerberos ticket — no password prompt at all.

- During setup, Entra Connect creates a special computer account called **AZUREADSSOACC** in your on-prem AD — this represents Entra ID itself, from AD's point of view, for Kerberos purposes.
- The user's browser silently requests a Kerberos ticket from DC01 for this account, then hands that ticket to Entra ID as proof of identity.
- This only works on domain-joined devices that are on the corporate network with line-of-sight to a domain controller.

Seamless SSO — No Password Prompt on Domain-Joined Devices



Requires: hybrid device on the corporate network, line of sight to DC01, and the AZUREADSSOACC computer object created during SSO setup

Diagram 8 — the 4-step silent Kerberos handshake behind Seamless SSO.

ENABLE SEAMLESS SSO

- 1 On AADCONNECT01 → re-run the Entra Connect wizard → Additional tasks → **Customize synchronization options** → sign in with Global Admin → on the Optional features page, check **Seamless single sign-on** → Next through to completion
- 2 This automatically creates the AZUREADSSOACC computer object in your on-prem AD (visible in Active Directory Users and Computers)

OPTIONAL: BUILD CLIENT01 & TEST

- 1 Create a Windows 10/11 VM named **CLIENT01** in subnet-management (10.10.2.10), same VNet so it can reach DC01 for DNS
- 2 Domain-join it to ad.minjtech.xyz the same way as AADCONNECT01 (System → Rename this PC → Domain)

- 3 Log in as **nishant@minjtech.xyz** using the password set earlier
- 4 Open a browser and go to a Microsoft cloud app (e.g. portal.office.com) — with Seamless SSO working, you should NOT be prompted for a password

```
whoami
whoami /user
whoami /groups
ipconfig /all
klist
```

- **whoami / whoami /user** — confirms which domain account is actually logged in.
- **whoami /groups** — shows AD group memberships applied to this session.
- **klist** — lists currently cached Kerberos tickets; after opening a cloud app with SSO working, you should see a ticket referencing AZUREADSSOACC.

Section 14: End-to-End Flow — Putting It All Together

This is the complete sequence, from an admin creating a user to that user opening a cloud app — every concept from this guide, in order.

End-to-End: From User Creation to Cloud App Access

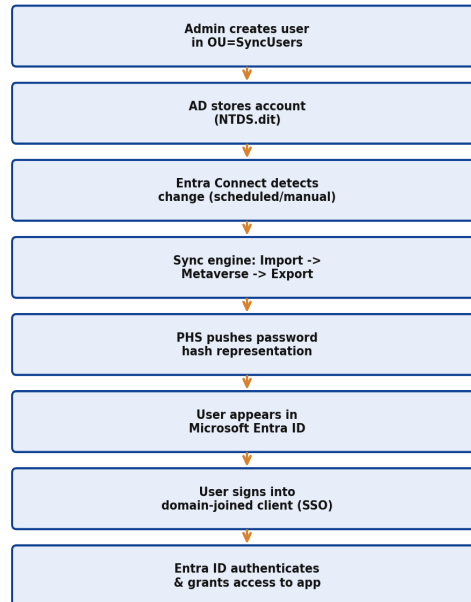


Diagram 9 — the full journey: user creation → AD storage → sync → PHS → Entra ID → SSO → app access.

Section 15: Monitoring Synchronization & Entra Connect Health

CONCEPT — In Simple Words

Sync runs quietly every 30 minutes by default — you need a way to CHECK it's actually succeeding, not just assume it is. Two tools: the local **Synchronization Service Manager** app on AADCONNECT01 (good for detailed, real-time troubleshooting) and cloud-based **Microsoft Entra Connect Health** (good for ongoing monitoring and alerts, viewable from anywhere).

LOCAL MONITORING — Synchronization Service Manager

- 1 On AADCONNECT01 → Start menu → **Synchronization Service** (installed alongside Entra Connect)
- 2 **Operations** tab → shows every run: Import, Synchronization (Delta/Full), Export — each with a status
- 3 Click any run to see counts of **Adds, Updates, Deletes**, and any errors
 - **Success** — completed with no errors.
 - **Warning** — completed, but with issues worth reviewing (e.g. a duplicate attribute).
 - **Error** — the run failed, usually needs immediate attention.
 - **Pending** — currently running or queued.

CLOUD MONITORING — Entra Connect Health

- 1 portal.azure.com → Microsoft Entra ID → left menu → **Microsoft Entra Connect** → **Connect Health** (or search "Azure AD Connect Health" directly)
- 2 View **Sync errors, Alerts, Performance** counters, **Last synchronization** time, and overall **Agent status** for AADCONNECT01

Section 16: Troubleshooting Lab — 6 Deliberate Failures

The best way to really learn hybrid identity is to BREAK it on purpose and fix it. Work through all 6 scenarios below on your lab environment.

FAILURE 1 — DNS Problem

- 1 Break it: on AADCONNECT01, temporarily change its DNS server setting away from 10.10.1.10 to something invalid
- 2 Observe: run **nslookup ad.minjtech.xyz** and **Resolve-DnsName ad.minjtech.xyz** — both fail to resolve
- 3 Fix: change AADCONNECT01's NIC DNS setting back to 10.10.1.10 (or confirm it's inheriting from the VNet)
- 4 Verify: re-run nslookup — should resolve to DC01 again

FAILURE 2 — Wrong UPN

- 1 Break it: create a test user with UPN **user@wrongdomain.local** instead of a real verified domain
- 2 Observe: this account can still sync, but will show as unverified/unusable for cloud sign-in — Entra ID cannot issue a usable identity for a domain it doesn't own
- 3 Fix: edit the user in ADUC → Account tab → change the UPN suffix dropdown to minjtech.xyz
- 4 Verify: run a delta sync, check the user in Entra ID shows the corrected UPN

FAILURE 3 — OU Filtering

- 1 Break it: move **testuser01** out of OU=SyncUsers into OU=ServiceAccounts (which is NOT in the sync scope)
- 2 Run sync: **Start-ADSyncSyncCycle -PolicyType Delta**
- 3 Observe: testuser01 disappears (or shows as "pending deletion"/gets removed) from Entra ID — because it's now outside the filtered OU scope
- 4 Fix: move testuser01 back into OU=SyncUsers → run sync again
- 5 Verify: confirm testuser01 reappears in Entra ID with the same Object ID as before (proving it's the SAME object, not a new one)

FAILURE 4 — Duplicate UPN / Attribute Conflict

- 1 Break it: create a second user with the exact same UPN as an existing synced user (e.g. two accounts both claiming nishant@minjtech.xyz)
- 2 Observe: sync will report an export/synchronization ERROR for a duplicate attribute value — one identity cannot map to two source objects
- 3 Fix: rename or delete the duplicate account so UPNs are unique again
- 4 Verify: re-run sync, confirm the error clears in Synchronization Service Manager

FAILURE 5 — Password Synchronization Issue

- 1 Symptom: a user's NEW on-prem password doesn't seem to work for cloud sign-in yet
- 2 Troubleshoot: PHS normally syncs password changes within 2 minutes via a separate, faster trigger (not the full 30-minute cycle) — first just wait a couple of minutes and retest

- 3 Check the event log on AADCONNECT01: Applications and Services Logs → Directory Synchronization — look for password sync-specific errors
- 4 Confirm the ADSync service itself is healthy (see Failure 6 below) — a stopped service blocks password sync too

FAILURE 6 — Synchronization Service Stopped

- 1 Break it: stop the sync service intentionally

```
Get-Service ADSync
Stop-Service ADSync
# ... make an AD change here, notice it never appears in Entra ID ...
Start-Service ADSync
Get-Service ADSync
```

- 1 Observe: while stopped, no scheduled sync runs at all — Get-ADSyncScheduler still shows a schedule, but nothing executes
- 2 Fix: **Start-Service ADSync**
- 3 Verify: run **Start-ADSyncSyncCycle -PolicyType Delta** manually and confirm the pending change now appears in Entra ID

Section 17: Staged Rollout & Security Design

STAGED ROLLOUT — CONCEPT

Traditional deployment turns on PHS or Seamless SSO for the ENTIRE organization at once — risky if something's misconfigured. **Staged rollout** lets you enable a new authentication feature for a small pilot SECURITY GROUP first, watch it work correctly for a few days, then gradually expand the group until everyone is covered — all without changing the actual federation/sync settings for the whole tenant.

- Configured in Entra ID → Microsoft Entra Connect → **Staged rollout** → enable PHS or Seamless SSO features for a specific pilot group.
- Real enterprises use this to de-risk migrations from ADFS to PHS, or when rolling out Seamless SSO gradually across departments.

SECURITY DESIGN — LAB vs PRODUCTION

- **Never expose DC01 directly to the Internet** — no public IP with open management ports. [LAB: acceptable to use temporary RDP with your own IP restricted in NSG for setup only. PRODUCTION: use Bastion, never direct RDP.]
- **RDP restrictions** — restrict NSG rules to your own IP only, remove after setup where possible.
- **Azure Bastion** — PRODUCTION RECOMMENDATION: use Bastion for all admin access instead of public RDP/SSH.
- **Just-in-time (JIT) VM access** — PRODUCTION RECOMMENDATION: via Microsoft Defender for Cloud, opens management ports only when needed, for a limited time.
- **Least privilege** — don't use Domain Admin/Global Admin for daily tasks; use scoped roles (Hybrid Identity Administrator, User Administrator).
- **Domain Admin vs Enterprise Admin vs Global Administrator** — Domain Admin controls one AD domain; Enterprise Admin controls the whole forest; Global Administrator controls the whole Entra ID tenant. These are DIFFERENT systems — don't assume one implies the other.
- **MFA** — PRODUCTION RECOMMENDATION: enforce for all admin accounts at minimum, ideally all users.
- **Conditional Access** — PRODUCTION RECOMMENDATION: enforce device compliance, location, and risk-based sign-in policies on top of basic authentication.
- **Privileged Identity Management (PIM)** — PRODUCTION RECOMMENDATION: makes admin roles time-bound/just-in-time instead of standing/permanent.
- **Service accounts** — keep in a dedicated OU (ServiceAccounts, excluded from sync in our lab), use strong non-expiring-but-rotated passwords or Managed Identities where possible.
- **Backup & Domain Controller recovery** — PRODUCTION RECOMMENDATION: regular System State backups of DC01, documented DSRM password storage, and a tested recovery runbook.
- **Azure VM security** — keep DC01/AADCONNECT01 patched, enable Azure Defender for Servers, restrict NSGs to only required ports between the two servers.

Section 18: Complete Port Matrix

Required connectivity between AADCONNECT01, DC01, and clients. Use this to build correct NSG rules.

Port	Protocol	Source → Destination	Purpose	Required?
53	TCP/UDP	AADCONNECT01/Client → DC01	DNS lookups	Yes
88	TCP/UDP	Client/AADCONNECT01 → DC01	Kerberos authentication	Yes
135	TCP	AADCONNECT01 → DC01	RPC endpoint mapper	Yes
389	TCP/UDP	AADCONNECT01 → DC01	LDAP	Yes
445	TCP	AADCONNECT01/Client → DC01	SMB (SYSVOL/NETLOGON access)	Yes
464	TCP/UDP	AADCONNECT01/Client → DC01	Kerberos password change	Yes
636	TCP	AADCONNECT01 → DC01	LDAPS (encrypted LDAP)	Recommended
3268	TCP	AADCONNECT01 → DC01	Global Catalog (LDAP)	If multi-domain forest
3269	TCP	AADCONNECT01 → DC01	Global Catalog (LDAPS)	If multi-domain forest
443	TCP	AADCONNECT01 → Internet (Entra ID)	HTTPS sync traffic to Entra ID	Yes

Azure networking implication: all DC01↔AADCONNECT01 ports above only need to be open WITHIN the VNet (same subnet or peered subnets) — never expose 389/636/88/53/445 etc. on a public IP or via an NSG rule allowing Internet (0.0.0.0/0) source. Only port 443 outbound to the internet (for Entra ID sync) is genuinely internet-facing, and that's outbound-only.

Section 19: Complete Lab Validation Checklist

- Azure VNet created (10.10.0.0/16)
- Both subnets created (subnet-domain, subnet-management)
- NSG configured and associated
- DC01 created with static private IP 10.10.1.10
- AD DS + DNS roles installed on DC01
- Forest ad.minjtech.xyz created
- dcdiag runs clean with no errors
- DNS SRV records (_ldap, _kerberos) visible in DNS Manager
- UPN suffix minjtech.xyz added
- OU structure (MinjTech > Users/Groups/Computers/ServiceAccounts/Admins/SyncUsers) created
- 4 test users created inside OU=SyncUsers with minjtech.xyz UPN
- 3 groups created and populated
- minjtech.xyz added as custom domain in Entra ID
- minjtech.xyz TXT record added in GoDaddy and verified in Entra ID
- AADCONNECT01 created with static private IP 10.10.1.20
- AADCONNECT01 domain-joined to ad.minjtech.xyz
- Connectivity to DC01 verified (ports 53/88/135/389/445/464/636)
- Microsoft Entra Connect installed with Customize path
- PHS selected and configured
- OU filtering scoped to Users/SyncUsers/Groups only
- Initial sync (Start-ADSyncSyncCycle -PolicyType Initial) completed successfully
- nishant@minjtech.xyz visible in Entra ID with On-premises sync = Yes
- Password sync verified (new on-prem password works for cloud sign-in)
- Seamless SSO enabled (AZUREADSSOACC computer object created)
- CLIENT01 domain-joined (optional)
- SSO verified — no password prompt on domain-joined client
- Synchronization Service Manager reviewed (Operations tab)
- Entra Connect Health reviewed in the portal
- All 6 troubleshooting scenarios completed and resolved

Section 20: Interview Questions & Answers

25+ questions covering everything in this lab, answered the way a real engineer would in an interview.

Q. What is Hybrid Identity and why do organizations use it?

A. It's keeping on-prem Active Directory as the identity source of truth while synchronizing accounts into Microsoft Entra ID, so users get ONE identity that works for both on-prem resources and cloud apps like Microsoft 365 — avoiding duplicate accounts and duplicate passwords.

Q. What's the difference between identity synchronization and authentication?

A. Synchronization is the one-way, scheduled COPYING of account data from AD to Entra ID (runs every ~30 min). Authentication is the real-time act of proving identity at sign-in — it happens on every login, independent of when the last sync ran.

Q. What does Microsoft Entra Connect actually do?

A. It's the sync engine bridging on-prem AD and Entra ID: it imports AD objects, merges them centrally in the Metaverse, and exports the result to Entra ID, plus it optionally handles PHS and Seamless SSO configuration.

Q. Explain PHS in your own words.

A. AD already stores a hash of each user's password, never the plaintext. Entra Connect reads that hash, applies an additional hash/salt for secure transport, and stores that in Entra ID. The plaintext password never leaves the on-prem network.

Q. When would you choose PTA over PHS?

A. When company policy strictly forbids any password-derived data leaving on-prem, or when you need to enforce on-prem sign-in restrictions (like disabled account checks) in real time rather than relying on periodic sync.

Q. When would you choose ADFS over PHS/PTA?

A. When you need complex custom authentication scenarios — smart card auth, third-party MFA integration at the federation layer, or strict compliance mandates that require full on-prem authentication control.

Q. What is OU filtering and why use it?

A. It restricts which Organizational Units get synchronized to Entra ID, so accounts like service accounts or computer objects that don't need a cloud identity are excluded — keeping the cloud directory clean and reducing licensing/security surface.

Q. A user exists in on-prem AD but is not appearing in Entra ID. How would you troubleshoot it?

A. First check if the user's OU is within the Entra Connect sync scope (OU filtering) — most common cause. Then check Synchronization Service Manager's Operations tab for export errors (e.g. duplicate attribute). Confirm the ADSync service is running. Force a delta sync with `Start-ADSyncSyncCycle -PolicyType Delta` and re-check.

Q. What is the Source Anchor / immutableId and why does it matter?

A. It's the permanent, unique value (usually the AD objectGUID) linking an on-prem object to its Entra ID counterpart. If you delete and recreate a user instead of renaming, you break this link and create a duplicate identity in the cloud.

Q. What's the difference between a Domain Admin, Enterprise Admin, and Global Administrator?

A. Domain Admin has full rights over one AD domain. Enterprise Admin has full rights across the whole AD forest (all domains). Global Administrator is an Entra ID role with full rights over the cloud tenant —

completely separate systems.

Q. Why should Entra Connect run on a separate server from the Domain Controller?

A. To minimize the DC's attack surface (fewer installed components = smaller risk) and to avoid any Entra Connect issue (crash, update, reboot) affecting core domain services like logins and DNS.

Q. What are SRV records and why does AD need them?

A. SRV records advertise WHERE a specific service lives (host + port + priority), such as `_ldap._tcp` or `_kerberos._tcp` — clients use these to automatically locate a domain controller without being hardcoded to one IP.

Q. Why shouldn't internal AD DNS records be published in public DNS (e.g. GoDaddy)?

A. It would expose internal infrastructure (domain controller names, service locations) to anyone on the internet, giving attackers a map of your internal network and inviting direct attacks on AD/Kerberos/LDAP ports.

Q. What is Seamless SSO and what makes it work?

A. It lets a user on a domain-joined, on-network Windows device sign into cloud apps without a password prompt, using a silent Kerberos ticket exchange against a special `AZUREADSSOACC` computer object created in AD during setup.

Q. What's the `AZUREADSSOACC` computer account for?

A. It represents Microsoft Entra ID as a Kerberos service principal inside your on-prem AD, so domain-joined clients can request a Kerberos ticket for Entra ID exactly like they would for any other domain resource.

Q. Why choose `ad.minjtech.xyz` as the AD domain instead of `minjtech.xyz` directly?

A. It keeps the internal AD namespace clearly separate from the internet-facing public domain, avoiding DNS resolution conflicts, while a UPN suffix still lets users log in with clean addresses like `nishant@minjtech.xyz`.

Q. Why is `minjtech.local` a bad choice for a new AD domain today?

A. It's a non-routable, non-real internet domain, which causes certificate issues (can't get a public cert for `.local`), complicates Entra Connect/UPN matching, and is no longer recommended by Microsoft.

Q. What is a UPN suffix and why add one?

A. It's the part after `@` in a login name, separate from the actual AD domain name. Adding `minjtech.xyz` as a suffix lets users log in with a clean email-style UPN even though the underlying AD domain is `ad.minjtech.xyz` — this also has to match for clean Entra ID sync.

Q. What is the Metaverse in Entra Connect terms?

A. The central, de-duplicated data store inside the sync engine where data from all connectors (AD, Entra ID) is merged and join rules are applied, before being exported back out to each connected system.

Q. How would you troubleshoot a DNS-related sync failure?

A. Run `nslookup/Resolve-DnsName` for the AD domain from `AADCONNECT01`, confirm the VM's DNS server setting points to the domain controller, check `Test-NetConnection` on port 53, and confirm the VNet's DNS settings weren't accidentally reverted to Azure-provided DNS.

Q. What happens if the `ADSync` service is stopped?

A. No scheduled or manual delta sync executes at all — AD changes (new users, password changes, group updates) simply stop flowing to Entra ID until the service is restarted and a sync cycle is triggered.

Q. What is Staged Rollout and why use it?

A. A way to enable a new authentication feature (PHS, PTA, or Seamless SSO) for a small pilot security group first, without changing settings for the whole tenant — lets you validate before a full rollout, reducing risk.

Q. What does Microsoft Entra Connect Health provide that local monitoring doesn't?

A. Cloud-based, always-available visibility into sync health, alerts, performance counters, and agent status — viewable from anywhere, without needing to RDP into AADCONNECT01, and useful for ongoing operational monitoring, not just point-in-time checks.

Q. Why is it important to never expose the Domain Controller directly to the Internet?

A. DC01 hosts extremely sensitive services (LDAP, Kerberos, DNS) — exposing those ports publicly invites direct attacks against your entire identity infrastructure. Access should go through Bastion/JIT, never a public IP with open management ports.

Q. What's the real difference between an OU and a Security Group?

A. An OU is an organizational container used for structure and Group Policy/sync-scope targeting — it is NOT a security boundary by itself. A Security Group is what actually grants or restricts access to resources.

Q. Scenario: password sync seems delayed for a user who just changed their password on-prem. What do you check?

A. Password changes use a faster, separate trigger (typically under 2 minutes) rather than the full 30-minute cycle — first just wait briefly and retest. Then check the ADSync service status, review Directory Synchronization event logs on AADCONNECT01, and confirm no export errors are blocking the pipeline.

End of Day 3 detailed guide. Every concept, diagram, portal navigation step, hands-on lab, troubleshooting scenario, and interview question above maps to the original Day 3 Hybrid Identity brief.